Mn Incorporation and Mechanical Activation Synergistically Boost the Oxygen Evolution Reaction (OER) Activity of Cobalt Tungstate

Andrea Griesi[a], Manjunath Chatti[b], Luca Saluta[b], Michele Ferri[b], Lea Pasquale[c], Dorwal Marchelli[d], Ambra Celotto[a,e], Francesco De Boni[c], Andrea Sala[f], Alexandre G. Brolo[g], Roman Krahne[d], Mauro Gemmi[f], Liberato Manna[b], Giorgio Divitini[a*]*

The oxygen evolution reaction (OER) is the main kinetic rate-determining step in water electrolysis, particularly in the acidic environments required for proton-exchange membrane (PEM) technologies. Developing acid-stable, Earth-abundant catalysts with low energy barriers is therefore crucial to overcoming sluggish reaction pathways and enabling efficient, large-scale hydrogen production. Herein, we report a series of $Mn_xCo_{(1-x)}WO_4$ ($0 \le x \le 1$) wolframite-type oxides as noble-metal-free electrocatalysts for acidic water oxidation. Comprehensive structural and spectroscopic analyses confirm the formation of a continuous solid solution resulting from isovalent substitution of Mn^{2+} for Co^{2+} at the Co site, while preserving the W^{6+} framework. Mechanical activation via ball milling enables uniform cation distribution and enhances electrochemically accessible surface area, leading to significantly improved OER performance. Among the compositions studied, $Mn_{0.25}Co_{0.75}WO_4$ exhibits the highest activity in 0.5 M H_2SO_4, delivering a low overpotential of ~0.42 V at 10 mA cm^{-2} along with favourable kinetics and good operational stability. This study establishes a clear relationship between composition, structure and activity, and highlights cation substitution combined with mechanical activation as an effective strategy for designing robust, acid-tolerant OER catalysts.

1. Introduction

The growing demand for renewable and sustainable energy sources has intensified global efforts to produce green hydrogen *via* water electrolysis, a process that can significantly contribute to achieving carbon neutrality[1–5]. This process aligns well with a net-zero strategy, as it can be powered by intermittent renewable energy sources, such as solar, tidal, and wind energy[1,6–8]. Water electrolysis involves two coupled half-reactions, namely the hydrogen evolution reaction (HER) and the oxygen evolution reaction (OER). Among these, the OER constitutes the primary kinetic bottleneck due to its sluggish reaction kinetics, arising from a complex multi-step mechanism involving multiple proton-coupled electron transfer processes and the formation of O-O bonds. The intrinsically slow kinetics of the OER necessitate large overpotentials to reach technologically relevant current densities, which in turn reduces the overall energy efficiency of water electrolysis systems[9–12]. The kinetics of the anodic OER require large overpotentials (η) above the theoretical thermodynamic value of 1.229 V to drive the reactions at practical rates[13–15]. This limitation fundamentally restricts the energy conversion efficiency of the overall process, posing a significant challenge for the large-scale implementation of water electrolysis systems[12,13]. In acidic electrolytes, the kinetic challenges of the OER are exacerbated by severe materials constraints. Although acidic operation is advantageous for proton-exchange membrane (PEM) electrolysers - enabling high current densities, compact system architectures, and efficient gas management - the highly corrosive environment severely limits the availability of stable, active anode materials[16,17]. Most Earth-abundant transition-metal oxides that perform well in alkaline media suffer from rapid dissolution or surface degradation under acidic OER conditions, resulting in poor durability[16]. Consequently, IrO_2 and RuO_2-based catalysts remain the benchmarks in acidic environments due to their comparatively high activity and corrosion resistance, despite their high cost and, in the case of RuO_2, limited long-term stability at anodic potentials. This intrinsic activity/stability/cost trade-off represents a major barrier to the large-scale deployment of PEM water electrolysis[16,18] and underscores the need for alternative OER catalysts based on Earth-abundant elements that combine high efficiency with robust acidic stability. Thus, recent research efforts have increasingly focused on transition-metal-based oxides (e.g., Ni, Co, Fe, and Mn), which offer a promising balance of competitive OER activity and enhanced stability under harsh electrochemical acidic OER conditions[16,18]. The rational design of such catalysts demands precise control over their electronic structure, conductivity, and active-site density to optimise the adsorption–desorption kinetics of OER intermediates. In this regard, Co-based materials have emerged as particularly attractive candidates for acidic water oxidation due to their versatile redox chemistry and tunable electronic structure, which can facilitate efficient OER pathways under strongly acidic environments[16,17,19]. While cobalt antimonates have been more extensively investigated, comparatively little attention has been directed toward cobalt tungstates, particularly with respect to compositional engineering and systematic structure–activity correlations under acidic OER conditions. This gap in understanding motivates further exploration of tungstate-based systems[20–22]. Cobalt tungstate ($CoWO_4$) has recently emerged as a promising candidate among tungstate-based materials, owing to its high chemical stability, robust crystalline structure, and tunable electronic environment[23,24]. Theoretical studies suggest that $CoWO_4$ can exhibit reasonable OER activity, with calculated overpotentials of 0.9–1.0 V[25]. Experimental reports have

confirmed that nanostructured $CoWO_4$ demonstrates bifunctional catalytic behaviour for both OER and ORR, particularly when nanocrystalline phases or amorphous domains are present[26–28]. Nevertheless, the intrinsic catalytic activity of $CoWO_4$ remains moderate due to its limited electrical conductivity and relatively low density of exposed Co^{2+}/Co^{3+} redox sites[28]. A well-established strategy to enhance the catalytic performance of transition-metal oxides involves partial substitution with other 3d metals to modulate their electronic structure, defect chemistry and redox behaviour. Manganese has proven particularly effective at improving OER activity, mainly by optimising charge-transfer characteristics and stabilising catalytically relevant high-valence states. For these reasons, Mn was selected as the dopant for $CoWO_4$, as it provides an optimal balance between structural compatibility and electronic modulation[29]. Mn^{2+} shares the same oxidation state and octahedral coordination environment as Co^{2+}, enabling isovalent substitution within the wolframite lattice without requiring charge-compensation mechanisms or generating undesired secondary phases. Its slightly larger ionic radius induces a controlled monoclinic distortion and a gradual expansion of the unit cell, features that are fully consistent with the formation of a continuous $Mn_xCo_{(1-x)}WO_4$ solid solution. Beyond these structural considerations, Mn incorporation also modifies the local electronic environment in a favourable way by increasing redox accessibility, enhancing defect density and improving charge-transport pathways, all of which contribute to more efficient OER kinetics. These combined factors make Mn an ideal dopant for $CoWO_4$ and provide a clear rationale for its selection as a means to tailor both the structural and catalytic properties of the material[29]. In this work, we present a systematic investigation of the Mn-doped cobalt tungstate system $Mn_xCo_{(1-x)}WO_4$ ($0 \leq x \leq 1$), synthesised via a solid-state reaction followed by mechanical ball milling. The study aims to elucidate the structural, morphological, and electrochemical effects induced by Mn incorporation and mechanical processing on the acidic OER performance of $CoWO_4$. A combination of X-Ray Diffraction (XRD), micro-X-Ray Fluorescence (μ-XRF), X-ray Photoelectron Spectroscopy (XPS), and Scanning Electron Microscopy (SEM) analyses was employed to assess the degree of doping, elemental homogeneity, and microstructural evolution. Electrochemical characterisation, including linear sweep voltammetry, Tafel slope analysis, and chrono-potentiometric stability tests, was performed in acidic media to correlate catalytic activity with structural features. Overall, this work provides fundamental insights into the structure–property relationships governing OER activity in Mn-doped tungstates and proposes a scalable approach for designing cost-effective, robust catalysts for sustainable hydrogen production.

2. Experimental details

2.1 Materials

The precursors for the synthesis of materials have been purchased from Sigma-Aldrich without any further purification: manganese(II)oxide (MnO, 99%), cobalt oxide (CoO, >99.5%) and tungsten(VI) oxide (WO_3, 99%). All chemicals and solvents for electrochemical analysis were of analytical grade and used as received, without any additional purification. Platinised titanium fibre felt (Pt/Ti) was purchased from Fuel Cell Store (product code 592796-2).

2.2 X-ray Powder Diffraction (XRD)

XRD patterns were collected using a Malvern PANalytical Empyrean X-ray diffractometer equipped with a 1.8 kW sealed Cu Kα ceramic tube. The instrument features a high-resolution two-axis vertical goniometer (radius, 240 mm; angular resolution, 0.0001°) and a programmable XYZ sample stage for precise alignment. Data were collected using a Pixcel3D 2×2 area detector 28.4 × 28.4 mm, 516 × 516 pixels) operating at 45 kV and 40 mA. The diffraction patterns were collected at ambient conditions in a parallel-beam geometry and symmetric reflection mode over an angular range of 8°- 90°, with a step size of 0.026°. The High Score 4.1 software from PANalytical was used for preliminary phase identification, and the Rietveld refinement was performed using the Jana2020 software[30]. To measure the samples, all powders without any other manipulation were placed on a zero-diffraction silicon wafer.

2.3 Scanning Electron Microscopy (SEM) coupled with Energy-Dispersive X-ray spectroscopy (EDX)

Morphological analyses were carried out using a ZEISS Gemini SEM 560 equipped with a Schottky field-emission source. The sample powders were dispersed in water and deposited onto silicon chips. The images of the samples were collected at an accelerating voltage of 5 kV. Elemental analysis and mapping were performed using an Oxford XMax-80 silicon drift detector (80 mm^2 SDD) for energy-dispersive X-ray spectroscopy.

2.4 Micro-X-ray fluorescence (μ-XRF)

Micro X-ray fluorescence measurements were performed with a M4 Tornado spectrometer (Bruker, Germany). The system is equipped with a Rh-anode X-ray tube cooled by air and operated at 50 kV and 199 μA. Poly-capillary optics are used to focus the polychromatic beam to a spot size of 25 μm or less for Mo K_α. Detection of fluorescence radiation is performed by an energy-dispersive silicon drift detector with 30 mm^2 sensitive area and an energy resolution of <145 eV for Mn K_α. Measurements were carried out directly on the pellet samples placed on the instrument platform in a low-vacuum environment (20 mbar). Quantification was performed using a standardless fundamental parameters (FP) method implemented in the Esprit software by Bruker. Each reported value is the average of at least five measurements taken at different points on the same sample.

2.5 X-ray Photoelectron Spectroscopy (XPS)

XPS measurements were carried out using a Kratos Axis UltraDLD spectrometer (Kratos Analytical Ltd.) with a monochromated Al K_α X-ray source (hν = 1486.6 eV) operating at 20 mA and 15 kV. Each specimen was prepared by pressing about 20 μg of powder onto an indium substrate until a

flat, homogeneous film formed. The wide scans were collected over an analysis area of 300 × 700 µm^2 at a photo-electron pass energy of 160 eV and an energy step of 1 eV, while high-resolution spectra were collected at a photo-electron pass energy of 10 eV and an energy step of 0.1 eV, corresponding to an instrumental resolution of 0.3 eV. A take-off angle of 0° with respect to the sample normal direction was used for all analyses. The spectra have been referenced to the C-C component of the adventitious carbon 1s peak at 284.8 eV. The spectra were analysed with the CasaXPS software (Casa Software Ltd., version 2.3.2531), and the residual background was removed using the Shirley method across the full binding energy range of the peaks of interest.

2.6 Ball Milling

Ball milling was carried out in a MM400 supplied by InSolidoTechnologies. Zirconia (ZrO_2) jars were thoroughly cleaned to avoid any cross-contamination, then filled with 5 mm x Ø5 mm. The samples, carefully weighed, were milled at 15 Hz for 75 minutes with a ball-to-powder ratio (BPR) of 40. The fine powders obtained were carefully removed from all surfaces of both jars and balls to recover the maximum possible product.

2.7 Nitrogen Physisorption

The specific surface area and porosity of the samples, both before and after ball milling, were determined by nitrogen physisorption at 77 K using an automated gas sorption analyser (3Flex, Micromeritics, USA). Prior to measurements, the powder samples were degassed under vacuum at 200 °C overnight to remove any adsorbed species. The specific surface area was calculated using the multi-point Brunauer–Emmett–Teller (BET) method, based on equally spaced points in a relative pressure range (p/p_0) from 0.05 to 0.30, with a correlation coefficient of above 0.999. The pore-size distribution was derived by the Barrett–Joyner–Halenda (BJH) method. The uncertainty in determining the adsorbed nitrogen amount and the specific surface area was estimated at 5–10%, accounting for the instrument's systematic error and the limitations of the experimental setup.

2.8 Raman Spectroscopy

Raman spectra were acquired using a Renishaw inVia Raman microscope. The incident laser (at 532 nm and 50 mW, spot size 1 µm) was directed onto the sample through an Olympus ULWD MSPlan 50X microscope objective. The scattered light was collected through the same objective, selected by an edge filter, and directed to the spectrograph. The scattered light was then detected using a Renishaw CCD camera.

2.9 Electrochemical measurements

All electrochemical measurements were carried out using a BioLogic SP 300 electrochemical workstation in a three-electrode configuration. The OER electrocatalytic experiments were conducted in a two-compartment cell, where the working and counter electrode compartments were separated by a low-porosity P4 glass frit. Before use, all glassware was immersed overnight in freshly prepared aqua regia to eliminate any trace metal contaminants. A high-surface-area platinum wire auxiliary electrode was separated from the working electrode compartment in all experiments performed at ambient (23 ± 2 °C) or at elevated temperatures (50 ± 1 °C and 80 ± 1 °C). High-temperature conditions (measured using a high-precision mercury thermometer) were achieved using a sand bath, with sand procured from Sigma-Aldrich (product code 274739). All potentials were measured against either a $Hg|HgSO_4|K_2SO_4$(sat.) reference electrode or a reversible hydrogen electrode (RHE) (obtained from Gasketel). All the electrocatalytic tests were undertaken in 0.5 M H_2SO_4 electrolyte solution, and approximately 40 mL of 0.5 M H_2SO_4 electrolyte solution (pH 0.3, measured using a *sensION+* pH meter) was added to each compartment. The electrolyte was stirred at approximately 400 rpm using a Teflon-coated magnetic stir bar throughout the electrochemical measurements. For the electrocatalytic tests, Pt–Ti felt was used as the substrate. The Pt-Ti felt electrodes were initially treated with a concentrated H_2SO_4:H_2O_2 (30%) solution in a 3:1 volume ratio and thoroughly rinsed with water. The total active geometric area of the felt, including all sides, was 0.25 cm^2, carefully measured to ensure consistent normalisation of the electrochemical data. Catalyst suspensions were prepared by dispersing the catalytic material and Vulcan carbon in a 4:1 wt/wt ratio in 1 mL of a solvent mixture containing 0.70 mL water, 0.25 mL isopropanol, and 0.05 mL of 5 wt% Nafion ethanol dispersion as the binder. The suspensions were sonicated for 60 min (VWR Ultrasonics) to achieve uniform dispersion and immediately drop-cast onto the freshly cleaned Pt-Ti electrodes, resulting in a catalyst loading of 0.5 mg·cm^{-2}. The uncompensated resistance (R_u) was determined from electrochemical impedance spectra recorded over a frequency range of 200 kHz to 0.1 Hz with an amplitude of 10 mV at potentials where negligible faradaic processes occur (typically 0.02–0.07 V vs RHE). Typical R_u values for the Pt-Ti electrodes at pH 0.3 were within the range 3-5 Ω (23 ± 2 °C), 3 ± 1 Ω (50 ± 1 °C) and 2 ± 1 Ω (80 ± 1 °C). Unless specified, reported potentials were corrected for ohmic losses by manually subtracting the iR_u contribution. The Tafel plots reported in this work were constructed from iR_u-corrected galvanostatic polarisation data, and the corresponding correction was applied consistently for all catalyst compositions and temperatures investigated.

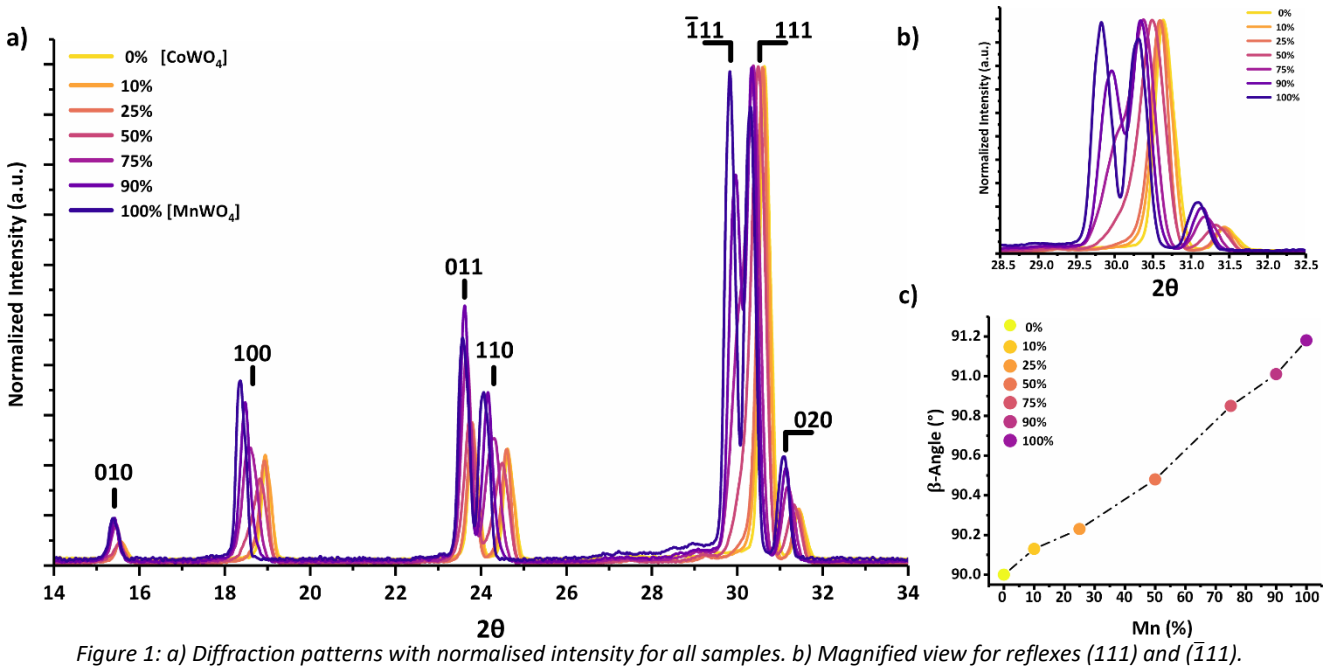


Figure 1: a) Diffraction patterns with normalised intensity for all samples. b) Magnified view for reflexes (111) and ($\bar{1}$11). c) Variation of β as the percentage of manganese increases.

Results and Discussion

The $Mn_xCo_{(1-x)}WO_4$ series (x = 0, 0.10, 0.25, 0.50, 0.75, 0.90, and 1.00) was synthesised via a conventional solid-state reaction route by modifying our previous method of synthesising cobalt tungstate[32] (see Experimental Section for details). Briefly, stoichiometric amounts of MnO, CoO, and WO_3 powders were accurately weighed and thoroughly mixed in an agate mortar to ensure homogeneous blending. The resulting mixtures were transferred to alumina crucibles and calcined in ambient air at 950 °C for 9 h using a controlled heating rate. Following calcination, the obtained solids were subjected to mechanical ball milling to reduce particle size and improve compositional uniformity.

2.10Powder X-ray diffraction

Powder X-ray diffraction (XRD) was employed to investigate the phase composition and crystal structure of the pre-ball-milled $Mn_xCo_{(1-x)}WO_4$ samples. Both $CoWO_4$ and $MnWO_4$ are known to crystallise in the monoclinic wolframite-type structure (Bravais lattice: monoclinic; space group P2/c)[33,34]. As shown in Figure 2a, the diffraction patterns of all compositions can be well indexed to this wolframite structure, with no detectable secondary or impurity phases, confirming the high phase purity of the synthesised materials. To further elucidate the structural evolution induced by Mn incorporation into the $CoWO_4$ lattice, detailed Rietveld refinements were subsequently performed to quantify changes in lattice parameters, unit-cell volume, and lattice distortion. Although $CoWO_4$ and $MnWO_4$ adopt the same wolframite-type framework, prior studies have identified a key geometric distinction between the two endmembers. Specifically, $CoWO_4$ exhibits a β angle of 90° despite its monoclinic symmetry, whereas $MnWO_4$ shows a β angle exceeding 90°, accompanied by a slightly larger unit-cell volume[35]. The wolframite structure is composed of chains of edge- and corner-sharing metal–oxygen octahedra extending along the *c* axis, in which each MeO_6 unit (Me = Co or Mn) shares corners with adjacent WO_6 octahedra to form a distorted octahedral network. Within this framework, variations in octahedral tilting along the chains control the degree of monoclinic distortion and are directly reflected in the magnitude of the β angle (Figure 1c). These established structural characteristics provide an essential reference for interpreting the evolution of lattice parameters observed upon Mn substitution in the present $Mn_xCo_{(1-x)}WO_4$ system. Consistent with this structural framework, powder XRD patterns (Figure 1) show a systematic shift of diffraction reflections toward lower 2ϑ values with increasing Mn content, indicative of lattice expansion arising from the substitution of Co^{2+} (r = 0.74 Å) by the slightly larger Mn^{2+} cation (r = 0.83 Å). Rietveld refinements of all compositions (Figures S1–S7, Tables S1 and S2) reveal a monotonic increase in lattice parameters *a*, *b*, *c*, and unit-cell volume across the series, confirming the formation of a continuous Mn–Co tungstate solid solution. In parallel, the β angle increases progressively from 90.00° for $CoWO_4$ toward values approaching 91.18° for $MnWO_4$ (ICSD entries 19760 and 15850), reflecting increasing monoclinic distortion with Mn incorporation. This trend is further corroborated by the gradual lifting of degeneracy between the (111) and ($\bar{1}$11) reflections: while these peaks coincide in $CoWO_4$ due to the elementary cell dimension, increasing Mn content results in a clear and quantifiable peak splitting as β deviates from 90°. The systematic evolution of peak positions and splitting

provides robust structural evidence for complete Mn incorporation and the absence of secondary phases.

2.11 Effect of Ball Milling on Compositional Homogeneity

µ-XRF analysis provided essential information on elemental distribution and compositional uniformity. Prior to ball milling, the pellets prepared for XRF measurements exhibited inhomogeneities (Figure 2b and c, and Tables S3 b,c,d), with distinct areas rich in tungstate ($Mn/CoWO_4$) or tungsten(VI) oxide (WO_3 precursor of the synthesises) in the samples containing 10%, 25%, and 50% Mn. These variations are attributed to the intrinsic limitations of manual mixing during the initial solid-state preparation. Because the crystalline structures of the reactants and the target wolframite phase are very similar, such compositional deviations remain invisible to powder XRD, which averages structural information over the entire diffracting volume. µ-XRF, therefore, plays a crucial role in revealing micro- to mesoscale chemical heterogeneity that bulk diffraction techniques cannot resolve. After ball milling, the µ-XRF elemental maps show a fully uniform distribution of Mn, Co, and W across all samples. This can also be seen optically: green areas, characteristic of WO_3, are visible before ball milling (Figure 2b) but disappear after it (Figure 2c). This homogenisation arises from the high-energy mechanical impacts occurring during milling, which repeatedly fracture and redistribute any residual precursor-rich regions that did not fully react during calcination. The continuous collision–shear action within the jar further promotes intimate mixing at the sub-micrometric scale, driving the powders towards a uniform stoichiometry. As a result, the elemental ratios measured by µ-XRF converge toward the theoretical Mn:Co:W compositions at all targeted replacement levels (Tables S3 a-g), demonstrating that ball milling is highly effective at eliminating local compositional gradients. This effect is particularly relevant in mixed-metal oxide systems such as $Mn_xCo_{(1-x)}WO_4$, where the solid-state reaction is diffusion-limited and incomplete homogenisation is a common issue even when powder XRD indicates a single-phase pattern. The improved compositional uniformity has significant implications for the electrochemical performance of the catalysts. Local fluctuations in Mn content directly influence the oxidation-state distribution, the density of redox-active sites, and the local electronic structure, all of which affect charge-transfer kinetics during the electrochemical acidic OER. By enforcing a uniform cation distribution, ball milling results in more reproducible electrocatalytic activity, more coherent Tafel behaviour, and a more reliable correlation between composition and performance compared to the pristine material.

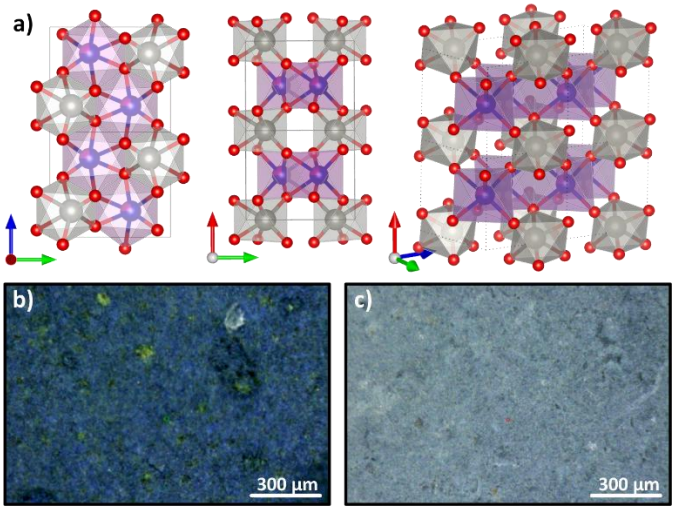


Figure 2: *a) Projection of the crystal structure of $Mn_{0.25}Co_{0.75}WO_4$. b-c) Optical images of the surface of $Mn_{0.25}Co_{0.75}WO_4$ pellets before and after ball milling.*

2.12 Scanning Electron Microscopy and Morphological Evolution

SEM was used to investigate the morphological and compositional evolution of the $Mn_xCo_{(1-x)}WO_4$ samples before and after mechanical ball milling. The as-synthesised powders obtained via solid-state reaction consist of relatively large particles with faceted surfaces, smooth edges and a broad size distribution, reflecting heterogeneous precursor mixing and the inherently slow diffusion kinetics associated with high-temperature solid-state synthesis. Following ball milling, a pronounced morphological transformation is observed across all compositions (Figure 3 and S15), with the large agglomerates fragmented into smaller particles, predominantly in the submicrometre-nanometre size range. The effect of the ball milling step is also reflected in the sample composition, as shown by the SEM-EDX maps before and after the treatment. The size reduction arises from repeated impact, shear, and abrasion during milling, which progressively break down the calcined particles. Concomitantly, the particles exhibit increased surface roughness and irregular edges, indicative of mechanical abrasion and the formation of a defect-rich surface.

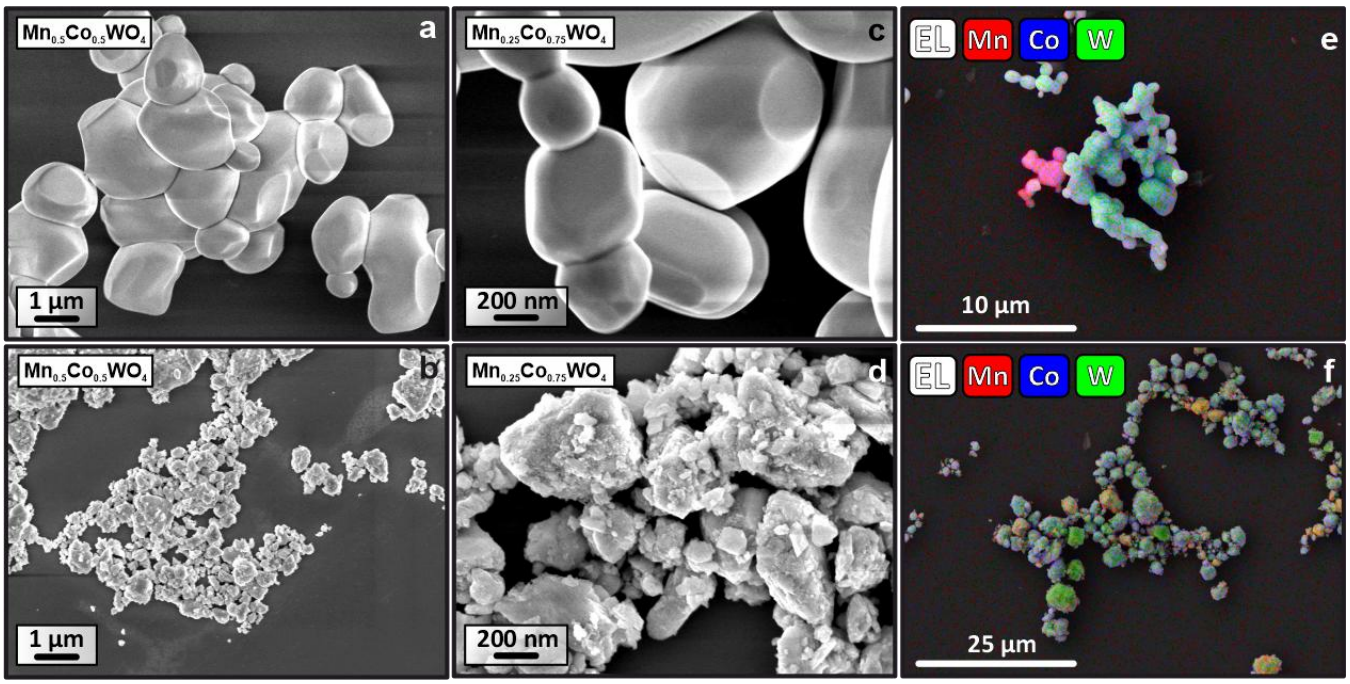


Figure 3: a-d) SEM images of samples $Mn_{0.5}Co_{0.5}WO_4$ and $Mn_{0.25}Co_{0.75}WO_4$ before (top) and after (bottom) ball milling. e,f) SEM-EDX elemental maps for $Mn_{0.25}Co_{0.75}WO_4$ before (e) and after (f) ball milling.

The reduced particle size is expected to increase the electrochemically accessible surface area, thereby promoting a more effective catalyst-electrolyte interaction. Moreover, ball milling improves catalyst dispersion and adhesion on the electrode substrate, facilitating reliable electrode fabrication. The mechanical stresses introduced during milling are also likely to induce micro-strain and surface defects within the wolframite lattice, features that can enhance charge-transfer properties and increase the density of surface redox-active sites, thereby contributing to improved electrocatalytic performance.

2.13Nitrogen Physisorption Analysis

The specific surface area and pore size distribution of the samples in $Mn_xCo_{(1-x)}WO_4$ series were determined by nitrogen adsorption. As shown in Figure 4, the N_2 adsorption-desorption isotherms of the ball-milled samples were consistent with a Type II isotherm according to the IUPAC classification[36], characteristic of macroporous adsorbents, in agreement with the electron microscopy observations discussed above. The hysteresis loops were of the H3 type[36], indicating a slit-shaped pore network composed of meso- and macropores, as further supported by the pore-size distribution curves shown in the inset of Figure 4. Similar values of specific surface area were obtained for all ball-milled samples (Table 1). By contrast, the data collected before ball milling, although not reported here, showed no identifiable monolayer formation, as typically observed for macroporous solids with weak adsorbent–adsorbate interactions. Therefore, the specific surface area could not be reliably determined.

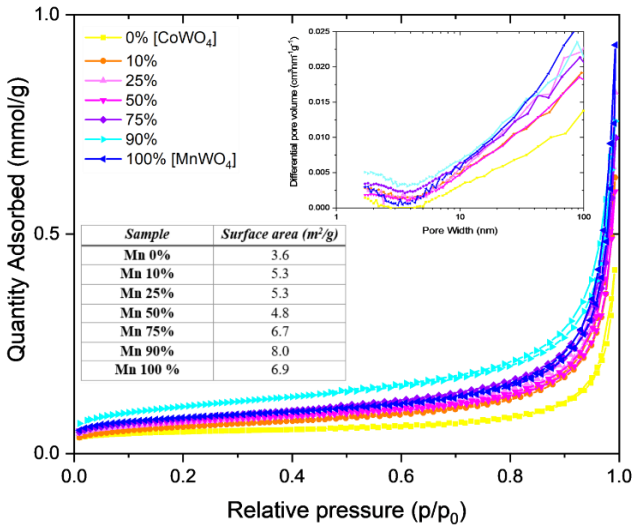


Figure 4: Nitrogen adsorption analysis results for the $Mn_xCo_{(1-x)}WO_4$ sample series after ball milling (left). Inset: pore size distribution curves calculated with the BJH method from the desorption branch of the isotherms (right), and Table 1 with specific surface area values (left).

2.14X-ray Photoelectron Spectroscopy

To further validate the successful synthesis of the $Mn_xCo_{(1-x)}WO_4$ compounds and to assess whether Mn and Co effectively compete for the same crystallographic site within the wolframite framework, XPS measurements were carried out on the representative composition $Mn_{0.25}Co_{0.75}WO_4$. XPS is particularly suitable for this purpose because it enables the

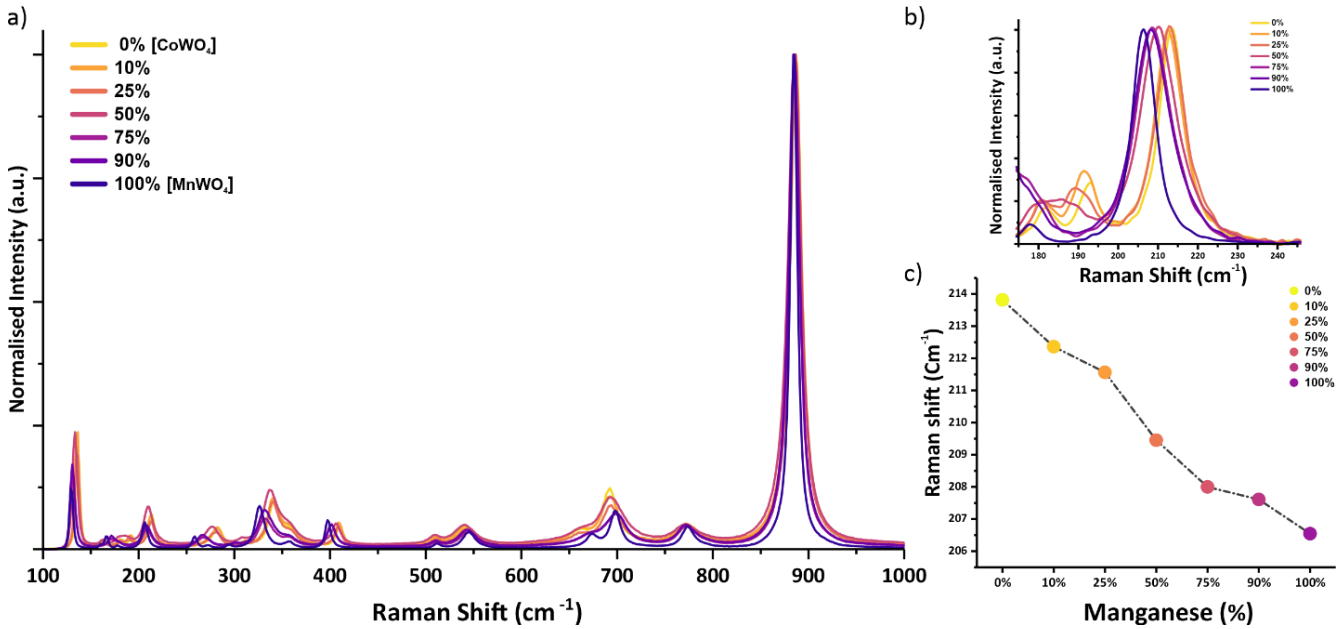


Figure 5: *a) Normalised Raman spectra of the $Mn_xCo_{(1-x)}WO_4$ series b) Magnified view of the spectra in the 175–250 cm^{-1} region, highlighting the Me–O stretching mode. c) Systematic red shift of the Me–O stretching-mode peak with increasing Mn content.*

determination of the oxidation states of transition metals on the surface – therefore the sites involved in catalysis – and provides insight into their local chemical environment.

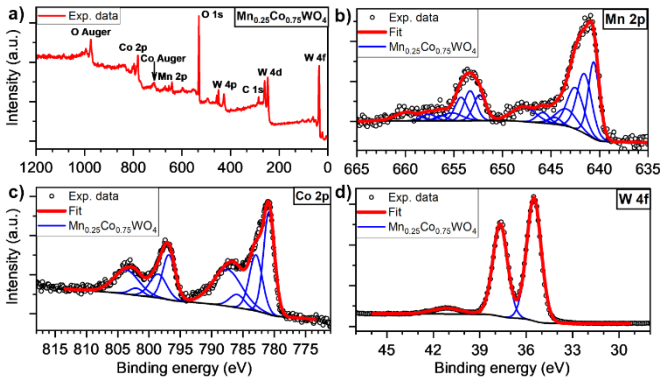


Figure 6: *a) Full XPS spectra of $Mn_{0.25}Co_{0.75}WO_4$. b to d) Magnified view of spectra for manganese, cobalt, and tungsten.*

For reference, the Mn 2p core-level spectra of $Mn_{0.25}Co_{0.75}WO_4$ were directly compared with those of two samples:

- Pure tungsten–manganese endmember $MnWO_4$
- Manganese precursor used in the synthesis, MnO.

This approach enables us to determine whether Mn in the doped sample retains its oxidation state when incorporated into the $CoWO_4$ lattice and to detect any potential changes in electronic configuration resulting from cation substitution or local structural distortions. The XPS Mn 2p region has a complicated multiplet splitting structure, requiring several components to fit the entire profile for a single oxidation state[36]. In this case, the theoretical spectral structure of MnO[36] was used to fit the Mn 2p region of all samples (Figure S10). As shown in Figure 6, the Mn $2p_{3/2}$ (first component centred at 640.5 eV) and Mn $2p_{1/2}$ (first component centred at 652.1 eV) peaks in the doped sample coincide almost perfectly with those of both $MnWO_4$ and MnO in terms of binding energy and overall spectral shape. The only slight difference is the presence of a further small satellite component centred at 647.8 eV for Mn $2p_{3/2}$ sub-region and 659.6 eV for Mn $2p_{1/2}$ sub-region in $Mn_{0.25}Co_{0.75}WO_4$ and $MnWO_4$ samples with respect to the MnO precursor. No additional peak shifts - typically indicative of mixed Mn^{2+}/Mn^{3+} valence states - are observed. This unequivocally demonstrates that Mn is present exclusively as Mn^{2+} in the sample, consistently with its expected coordination in the wolframite lattice. The absence of detectable oxidation-state changes upon doping further confirms that Mn substitutes directly into the Co^{2+} site without inducing cationic vacancies, charge compensation effects, or oxidation of tungsten. In other words, Mn doping does not alter the oxidation state of W^{6+}, and the stoichiometry of the wolframite structure remains preserved (Figure S9). This observation supports the structural interpretation derived from powder XRD and Rietveld refinements, confirming that the Mn atoms occupy the same crystallographic sites as Co and that the solid solution is homogeneous at the atomic level. Taken together, these XPS results reinforce the conclusion that Mn incorporation proceeds via a straightforward substitution mechanism at the divalent metal site in the wolframite structure, without altering the material's oxidation state. This behaviour is essential for ensuring predictable charge-transport properties and contributes to the consistent electrochemical performance observed in Mn-doped $CoWO_4$ catalysts[37,38].

2.15Raman Spectroscopy

As an additional verification of the successful incorporation of manganese into the cobalt sites of the wolframite lattice, Raman spectroscopy measurements were performed on the $Mn_xCo_{(1-x)}WO_4$ series. Both $MnWO_4$[39] and $CoWO_4$[40], (Figure 5) as well as their intermediate solid solutions, crystallise in the monoclinic wolframite structure (space group P2/c, point group C_2h), which contains two formula units per primitive cell. Group-theoretical considerations indicate that the vibrational representation of this structure is given by:

$$\Gamma_{(\text{Raman + infrared})} = 8A_g + 10B_g + 7A_u + 9B_u$$

where the even modes ($8A_g$ and $10B_g$) are Raman-active and the odd modes ($7A_u$ and $9B_u$) are IR-active[41,42]. The experimental spectra of pure phase (Fig. S12-13) exhibit Raman-active bands that align well with the expected symmetry-derived mode distribution. After assigning the Raman-active modes, particular attention was devoted to the stretching mode of the Me–O (Co/Mn–O) bond, which is highly sensitive to cation substitution within the wolframite framework. We found that a red shift of this stretching band is observed with increasing Mn content (Fig. 5). Although substitution of Co^{2+} by the lighter Mn^{2+} decreases the reduced mass of the Me–O pair, which alone would tend to increase the vibrational frequency, the observed red shift indicates that this mass effect is outweighed by a reduction in the effective force constant of the Me–O bond. This decrease in bond stiffness likely arises from weaker Mn–O bonding compared to Co–O, leading to a net lowering of the stretching-mode frequency. Raman spectra of $Mn_xCo_{(1-x)}WO_4$ show systematic shifts and broadening of the characteristic WO_6 vibrational modes with increasing Mn content, consistent with Mn incorporation into the $CoWO_4$ lattice and the introduction of substitutional disorder at the divalent metal site. The progressive shift of this Raman band therefore provides a sensitive vibrational signature of Mn incorporation, confirming that manganese substitution modifies the local bonding environment within the wolframite lattice. To quantify the decrease in the value of the binding constant between M-O as the amount of manganese increases, the Raman frequency shifts were correlated with the average bond lengths (*r*) derived from Rietveld refinement (see Fig. Sx). This confirms that the experimental red shift is governed by a reduction in the force constant (*k*) associated with the lattice expansion. This result is fully consistent with the structural evolution identified by XRD and the electronic configuration revealed by XPS, further supporting the formation of a coherent and continuous $Mn_xCo_{(1-x)}WO_4$ solid solution.

3. Electrochemical Discussion

The electrocatalytic performance of the synthesised cobalt tungstate and Mn-doped cobalt tungstate catalysts toward the OER was systematically evaluated in 0.5 M H_2SO_4 using Nafion as a binder. All electrochemical measurements were conducted under both ambient (23 ± 2 °C) and elevated

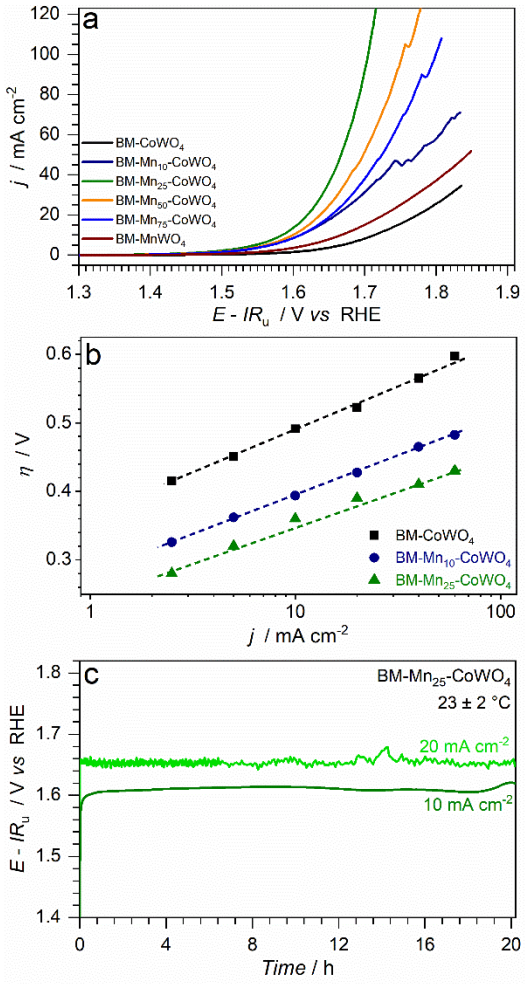


Figure 7: Electrochemical OER performance of $Mn_xCo_{(1-x)}WO_4$ after ball milling. a) Quasi-stabilised linear sweep voltammetry (scan rate 0.005 V s^{-1}; 5th scans shown); b) Polarisation plots derived from short-term chrono-potentiometric measurements shown in Figure S20; c) long-term durability tests of $Mn_{0.25}Co_{0.75}WO_4$ measured in galvanostatic mode at 10 mA cm^{-2} and 20 mA cm^{-2}.

temperatures (50 ± 1 °C and 80 ± 1 °C) to investigate the temperature-dependent activity and stability of the catalysts under acidic conditions. The initial OER performance of the pre-ball-milled catalytic materials was first examined. As shown in Figure S16, undoped $CoWO_4$ exhibited moderate OER activity, requiring an overpotential of approximately 570 mV to reach a current density of 10 mA cm^{-2}. Notably, this performance is inferior to that reported by Ranit et al.[44] (472 mV at 10 mA cm^{-2}), and we attribute this mainly to the relatively larger particle size and higher crystallinity of the

$CoWO_4$ samples synthesised in this work (Figure 3). Thus, these structural features limit the exposure of catalytically active surface sites and reduce the density of defect-rich regions, both of which are known to play a critical role in enhancing OER activity. Remarkably, Mn incorporation led to an improvement in OER performance. The overpotential decreased progressively with increasing Mn content up to 25%, reaching a minimum of 420 mV (1.65 V vs RHE). This enhancement is attributed to the synergistic electronic and structural effects induced by Mn partial substitution at Co sites, which improve the intrinsic catalytic activity. However, higher Mn loadings beyond this composition led to a slight increase in overpotential (1.66–1.67 V vs RHE for $Mn_{0.5}Co_{0.5}WO_4$ and $Mn_{0.75}Co_{0.25}WO_4$), likely due to excessive substitution that disrupted the optimal electronic configuration and reduced the density of active Co^{2+}/Co^{3+} redox sites. Notably, pure $MnWO_4$ exhibited poor OER activity (1.73 V vs RHE), further confirming that a balanced Mn–Co composition is critical for efficient OER performance. Overall, moderate Mn doping (25%) markedly enhances the OER activity of $CoWO_4$, highlighting the importance of optimised cation substitution to achieve superior catalytic efficiency in acidic media. We then tested the ball-milled samples for OER activity in 0.5 M H_2SO_4 solution. Linear sweep voltammetry (LSV) measurements (Figures 6a and S17) revealed that the ball-milled catalysts exhibited markedly improved electrochemical performance compared to their pre-ball-milled counterparts across all Mn doping levels. Notably, the same compositional trend in OER activity was observed, with the $Mn_{0.25}Co_{0.75}WO_4$ after ball milling sample again showing the highest activity among the series. This enhancement in catalytic performance is mainly attributed to two key factors. (i) The mechanical milling process has led to a substantial reduction in particle size, thereby increasing the electrochemically active surface area and facilitating greater access to catalytic sites, as evidenced by the SEM images shown in Fig 3 and surface area values found by the Nitrogen Physisorption Analysis (Figure 4). Additionally, the noticeable broadening of the XRD peaks suggests a partial loss of crystallinity and an average reduction in crystal size, which are known to promote catalytic activity by introducing more defect sites (Figure S8), and (ii) the ball milling also resulted in a significantly more homogeneous distribution of Mn throughout the $CoWO_4$. This uniform dopant distribution enhances charge transport and optimises the electronic environment at the active sites, thereby improving overall electrochemical performance. Most importantly, this observation confirms the key role of Mn incorporation and its homogeneous distribution in governing OER activity in acidic media (Figure 7a, S17). To further evaluate the contribution of surface area effects to the observed OER activity, we next undertook double-layer capacitance (C_{dl}) measurements for the ball-milled $Mn_xCo_{1-x}WO_4$ catalyst series using cyclic voltammetry in a non-faradaic potential region (Figures S18a). The C_{dl} values were determined from the slope of the linear relationship between the average capacitive current and scan rate (Figures S18b, Table S4). For the sake of completeness, it should be noted that C_{dl} measurements are not, per se, a direct measure of the real electrochemically active surface area (ECSA). As evidenced by Trasatti and Petrii in their milestone paper[45] on surface area measurements in electrochemistry, the double-layer capacitance is more representative of the electrochemically accessible and electrically conductive surface area than of the true catalytically active area participating in the reaction. Nevertheless, C_{dl} values provide a useful basis for internal comparison among catalysts of similar composition and morphology, such as those investigated in the present study. Therefore, capacitance ratios were calculated by normalizing the C_{dl} value of each catalyst to that of BM-$CoWO_4$, which exhibited the lowest capacitance within the ball-milled series (see Table S4). These ratios were subsequently used as normalization factors to correct the measured current densities, yielding the Cdl-normalised polarisation curves shown in Figure S19. Notably, the normalised polarisation curves retain essentially the same activity trend observed for the geometric-area-normalized data, with BM-$Mn_{0.25}Co_{0.75}WO_4$ remaining the most active catalyst in the series. This result indicates that the enhanced OER activity can not be solely attributed to the increased electrochemically accessible surface area associated with ball milling. Instead, the data suggest that Mn incorporation contributes to improving the intrinsic catalytic properties of cobalt tungstate, acting in concert with the increased accessible surface area generated by mechanical activation. Accordingly, all subsequent electrochemical analyses (*vide infra*) in this study were carried out exclusively on the $Mn_{0.25}Co_{0.75}WO_4$ ball-milled samples. Galvanostatic studies were then undertaken using short-term chrono-potentiometric (CP) measurements to evaluate the steady-state OER behaviour of the ball-milled catalysts. When compared in steady-state polarisation plots (Figures 7b and S16), the $Mn_{0.25}Co_{0.75}WO_4$ ball-milled catalyst again exhibited the highest activity among the series (Figure S17), consistent with the trends observed in LSV measurements. The reaction kinetics of the OER were further analysed by constructing Tafel plots from the quasi-steady-state polarisation data (Figure S16). The ball-milled $CoWO_4$ catalyst exhibited a Tafel slope of 0.117 V dec^{-1}, which decreased slightly to 0.113 V dec^{-1} and 0.106 V dec^{-1} for the $Mn_{0.10}Co_{0.90}WO_4$ and $Mn_{0.25}Co_{0.75}WO_4$ ball-milled samples, respectively (Figure S16 b). The gradual reduction in Tafel slope with increasing Mn content indicates more favourable OER kinetics and improved charge-transfer characteristics for the Mn-doped systems. These values are in good agreement with previously reported Tafel slopes (0.08–0.12 V dec^{-1}) for Co- and Mn-based oxides operating under acidic conditions, confirming the efficient catalytic nature of the present materials. The similar slope magnitudes among the samples suggest that the OER follows an analogous rate-determining step across the series, likely associated with the electrochemical O–O bond formation via proton-coupled electron transfer, as commonly observed for Co-based oxides

under acidic conditions. Furthermore, the relatively low Tafel slope (0.106 V dec^{-1}) obtained for $Mn_{0.25}Co_{0.75}WO_4$ after ball milling reflects its superior intrinsic catalytic kinetics, approaching the behaviour of benchmark OER catalysts such as IrO_2, thereby highlighting the beneficial role of optimised Mn incorporation in enhancing the overall reaction pathway. The long-term stability of the catalysts was subsequently evaluated by galvanostatic measurements for 20 h at room temperature under applied current densities of 10 and 20 mA cm^{-2} (Figure 7c). Notably, the overpotential required to achieve a current density of 10 mA cm^{-2} was as low as 0.40 ± 0.01 V, indicating highly efficient OER kinetics for the optimised catalyst. The $Mn_{0.25}Co_{0.75}WO_4$ ball-milled sample maintained a nearly constant potential throughout the measurement period at both current densities, demonstrating excellent durability under continuous OER

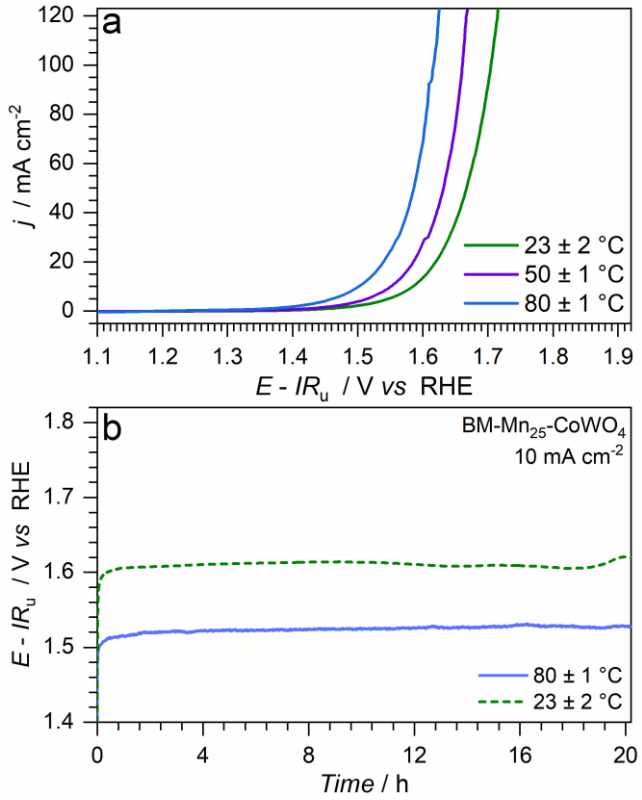


Figure 8: Temperature effect on OER performance of $Mn_xCo_{(1-x)}WO_4$ after ball milling. a) Quasi-stabilised linear sweep voltammetry (scan rate 0.005 V s^{-1}; 5th scans shown); b) long-term durability tests of $Mn_{0.25}Co_{0.75}WO_4$ measured in galvanostatic mode at 10 mA cm^{-2}.

operation. This sustained performance confirms that the superior activity observed in LSV measurements is not accompanied by rapid deactivation, highlighting that the optimised Mn doping provides both enhanced catalytic kinetics and robust structural stability in acidic media. Further, the effect of temperature on the OER catalysed by the ball-milled Mn-doped CoWO catalysts was systematically investigated at 50 ± 1 °C and 80 ± 1 °C. Linear-sweep voltammetry and short-term chrono-potentiometric measurements demonstrated the expected enhancement in catalytic performance with increasing temperature, as reflected in higher current densities at a given overpotential (Figures 8a and S17a). Tafel analysis revealed a slight decrease in slope with increasing temperature (c.a. 0.104 V dec^{-1}), indicating accelerated reaction kinetics and improved charge-transfer efficiency at elevated temperatures. The nearly linear η versus log j relationship observed at all temperatures suggests minimal mass- or charge-transfer limitations during the short-term measurements. The long-term stability of the $Mn_{0.25}Co_{0.75}WO_4$ ball-milled sample was subsequently evaluated at 80 °C under a constant current density of 10 mA cm^{-2} (Figure 8b). The catalyst maintained a nearly constant potential over the entire 20 h measurement period, demonstrating robustness under continuous OER operation at elevated temperature. Collectively, these results confirm that the Mn-doped $CoWO_4$ catalysts not only display enhanced catalytic activity at higher temperatures but also retain structural robustness and sustained performance under prolonged electrolysis in acidic media.

3.1 Post-reaction characterisation

To evaluate the structural stability of the catalyst and investigate potential phase transitions or material dissolution during operation, post-OER XRD, SEM and SEM-EDX analyses were performed (Figure 9 – S21). The catalyst was characterised after a long-term durability test conducted at 80 °C for 20 consecutive hours under acidic OER conditions. SEM imaging reveals partial loss or delamination of the catalyst layer from the substrate surface, a phenomenon frequently observed at harsh anodic potentials and high gas evolution rates; however, the initial morphology is still partly preserved. Furthermore, the elemental maps obtained by SEM-EDX confirm the uniform and persistent distribution of the catalyst components. Most importantly, when correlated with the post-mortem XRD data, it is evident that the ball-milled $Mn_{0.25}Co_{0.75}WO_4$ catalyst preserves its crystalline phase integrity. No secondary phases or structural degradation resulting from potential oxidation-state changes in cobalt or manganese were detected. This confirms that the bulk framework of the catalyst remains structurally robust during long-term acidic OER operation

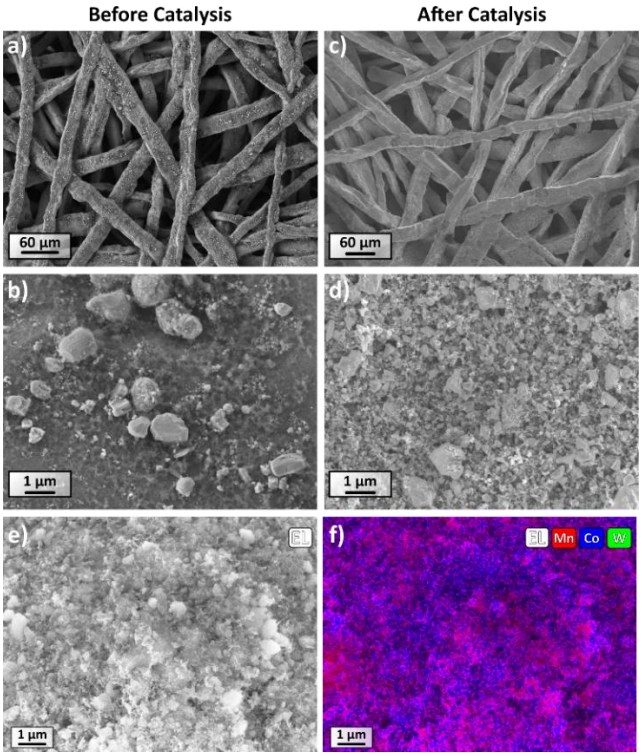


Figure 9: SEM images of the ball-milled $Mn_{0.25}Co_{0.75}WO_4$ electrode surface before (a, b) and after (c, d) the long-term durability test. e-f) *SEM-EDX elemental maps for $Mn_{0.25}Co_{0.75}WO_4$* after the test.

4. Conclusion

In this work, we synthesised and systematically investigated the $Mn_xCo_{(1-x)}WO_4$ solid-solution series to understand how controlled Mn incorporation and post-synthetic mechanical activation influence the structural, chemical, and electrocatalytic properties of cobalt tungstate for the oxygen evolution reaction in acidic media. XRD and Rietveld analyses demonstrated that Mn and Co form a continuous wolframite-type solid solution, with a progressive expansion of the unit cell and a systematic increase in the β angle—features that directly reflect the accommodation of the larger Mn^{2+} cation within the divalent metal site of the wolframite lattice. The clear splitting of the (111)/($\bar{1}$11) reflections confirmed homogeneous cation substitution and the absence of secondary phases across the entire compositional range. µ-XRF and SEM-EDX analysis revealed that solid-state synthesis alone produces incomplete mixing and local compositional fluctuations. However, these inhomogeneities were eliminated after ball milling, which induced sub-micrometre homogenisation. SEM observations corroborated this effect, showing a drastic reduction in particle size and the formation of roughened, defect-rich surfaces following milling. As further supported by the Nitrogen Physisorption Analysis, these structural and morphological modifications contribute to a more accessible surface area, improved site availability, and enhanced charge-transfer characteristics. XPS measurements confirmed that Mn is incorporated as Mn^{2+} without altering the oxidation state of tungsten or generating mixed-valence Mn species. Electrochemical measurements revealed that moderate Mn incorporation ($x \approx 0.25$) markedly enhances the OER performance of $CoWO_4$, lowering the overpotential to 420 mV at 10 mA cm^{-2} and improving Tafel slopes relative to undoped $CoWO_4$. Ball milling amplified this effect across the entire compositional series, with the $Mn_{0.25}Co_{0.75}WO_4$ sample consistently achieving the best activity, stability, and charge-transfer kinetics. The improved performance can be attributed to the synergistic combination of homogeneous Mn distribution, defect-rich microstructures, reduced crystallite size, and optimal tuning of the local electronic environment. Importantly, the ball-milled $Mn_{0.25}Co_{0.75}WO_4$ catalyst also demonstrated excellent long-term durability and consistently high OER activity over a temperature range from 25 °C to 80 °C, highlighting its robustness under harsh acidic and thermal conditions. Overall, this study provides a comprehensive structure–property–activity relationship for Mn-doped $CoWO_4$ and establishes mechanical activation as a powerful strategy for enhancing both compositional uniformity and catalytic performance. The results identify $Mn_{0.25}Co_{0.75}WO_4$ as a promising acid-stable, Earth-abundant OER catalyst and offer a scalable route for the rational design of tungstate-based mixed-metal oxides for sustainable hydrogen production.

Author contributions

A.G.: experiment design, sample synthesis, XRD data collection, XRD, SEM and Raman data processing, writing and paper organisation. M.C.: electrode preparation, electrochemical measurements and processing, writing. L.S.: electrode preparation, electrochemical measurements. M.F.: data processing and writing. L.P.: XRF data collection and processing, nitrogen absorption data processing, and writing. D.M.: Raman data collection. A.C.: SEM and EDX-SEM imaging. F.D.: XRF data collection, data processing and writing. A.S.: sample grinding. A.B. and R.K.: writing and supervising the Raman spectroscopy discussion. M.G.: writing and supervision of XRD data. L.M.: writing and supervision of electrochemical measurements. G.D.: writing and project supervision.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

D.M. and R.K. acknowledge funding by the European Union under Project 101131111 – DELIGHT. M.C. acknowledge funding from the EPNRRPE1 NEST Network 4 Energy Sustainable Transition.

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